

Project Documentation - ShopEZ

1. Introduction

Project Title: ShopEZ - E-commerce Application

Team Members: Kalanidhi Anushka (Individual Project)

2. Project Overview

Purpose: ShopEZ is a full stack MERN e-commerce web application that allows users to browse products by category, add items to cart and place orders. Admins can manage products and view all orders from a separate dashboard.

Features: User registration and login, product catalog with category filters and search, cart management, order placement, admin dashboard for product and order management.

3. Architecture

Frontend: Built with React.js and Tailwind CSS. Pages include Login, Register, Home, Cart, Orders and Admin. Axios is used for API calls.

Backend: Built with Node.js and Express.js running on port 5000. Routes handle auth, products, cart and orders. JWT and bcryptjs are used for authentication.

Database: MongoDB with Mongoose. Four collections - users, products, carts and orders.

4. Setup Instructions

Prerequisites: Node.js, MongoDB, Git

Installation:

- Clone the repo from github.com/kalanidhianushka15/shopez
- cd server, run npm install, then node index.js
- cd client, run npm install, then npm start
- Open <http://localhost:3000> in browser

5. Folder Structure

Client:

- src/pages - Login.js, Register.js, Home.js, Cart.js, Orders.js, Admin.js
- src/App.js - routing configuration

Server:

- models - User.js, Product.js, Cart.js, Order.js
- routes - auth.js, products.js, cart.js, orders.js
- index.js - main server file

6. Running the Application

Frontend: cd client, npm start (runs on localhost:3000)

Backend: cd server, node index.js (runs on localhost:5000)

7. API Documentation

- POST /api/auth/register - Register a new user
- POST /api/auth/login - Login and get JWT token
- GET /api/products - Get all products
- POST /api/products - Add a new product (admin)
- PUT /api/products/:id - Edit a product (admin)
- DELETE /api/products/:id - Delete a product (admin)
- POST /api/cart/add - Add item to cart
- GET /api/cart/:userId - Get user cart
- POST /api/orders - Place an order
- GET /api/orders/:userId - Get user orders

8. Authentication

JWT tokens are used for authentication. When a user logs in, the server generates a JWT token which is stored in localStorage. Passwords are hashed using bcryptjs before storing in MongoDB.

Admin routes are protected and only accessible if the user role is admin.

9. User Interface

The UI is built with React.js and Tailwind CSS. It has a sticky navbar, category filter buttons, product cards with images and prices, add to cart and buy now buttons, and a separate admin dashboard. The design is clean and responsive.

10. Testing

Manual testing was done for all features including registration, login, product browsing, category filters, search, add to cart and admin dashboard. All test cases passed successfully as documented in the UAT template.

11. Screenshots or Demo

GitHub Repository: github.com/kalanidhianushka15/shopez

Admin credentials: admin@shopez.com / admin123

User: Register a new account on the signup page

12. Known Issues

- Product images are loaded from external Unsplash URLs which may occasionally be slow to load
- No payment gateway integrated yet
- App runs locally and is not deployed to a cloud server

13. Future Enhancements

- Add payment gateway integration
- Add product reviews and ratings by users
- Add wishlist feature
- Add email notifications for order confirmation
- Add search suggestions and filters by price range